

## LANGLEY AFB, VA, USA

WMO: 745980

Lat: 37.083N

Lon: 76.360W

Elev: 3

StdP: 101.29

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 13702

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -7.0       | -4.6 | -17.3                         | 0.8 | -3.2 | -14.5 | 1.1 | -1.6 | 11.9                  | 5.5  | 10.8 | 6.7  | 4.0                   | 330  | 0.488 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 8.4                    | 33.6            | 24.6 | 32.4 | 24.3 | 31.3 | 23.9 | 26.4                | 30.6 | 25.8 | 29.8 | 25.2 | 29.1 | 4.6                  | 240  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 25.3                            | 20.4 | 27.9 | 24.8 | 19.8 | 27.6 | 24.1 | 19.0 | 27.1 | 82.0          | 30.4 | 79.2 | 30.0 | 76.8 | 29.0 | 30.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| 10.7              | 9.1  | 8.2 | -10.1                      | 35.8 | 3.0                | 1.5 | -12.3                                              | 36.8 | -14.1      | 37.7 | -15.8      | 38.5 | -18.0      | 39.6 |    |    |
|                   |      |     | -11.2                      | 27.5 | 2.8                | 1.0 | -13.2                                              | 28.2 | -14.8      | 28.8 | -16.4      | 29.3 | -18.4      | 30.0 |    |    |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 15.6   | 4.2   | 6.1   | 9.5   | 15.0  | 19.6  | 24.1  | 26.2  | 25.3  | 22.5  | 16.6  | 10.7  | 7.0  |
|                                                                                  | DBStd       | 8.79   | 5.81  | 5.16  | 5.21  | 4.84  | 4.15  | 3.23  | 2.54  | 2.43  | 3.08  | 4.30  | 4.61  | 5.22 |
|                                                                                  | HDD10.0     | 583    | 196   | 132   | 76    | 8     | 0     | 0     | 0     | 0     | 0     | 3     | 44    | 124  |
|                                                                                  | HDD18.3     | 1897   | 437   | 344   | 278   | 125   | 37    | 2     | 0     | 0     | 4     | 85    | 231   | 352  |
|                                                                                  | CDD10.0     | 2636   | 18    | 22    | 59    | 158   | 297   | 423   | 502   | 476   | 375   | 208   | 67    | 31   |
|                                                                                  | CDD18.3     | 907    | 1     | 1     | 3     | 24    | 76    | 176   | 244   | 217   | 129   | 32    | 3     | 1    |
|                                                                                  | CDH23.3     | 7658   | 1     | 4     | 24    | 195   | 552   | 1540  | 2427  | 1917  | 814   | 171   | 10    | 3    |
|                                                                                  | CDH26.7     | 2533   | 0     | 0     | 2     | 42    | 156   | 549   | 937   | 621   | 194   | 31    | 1     | 0    |
| Wind                                                                             | WSAvg       | 4.1    | 4.3   | 4.5   | 4.8   | 4.7   | 4.2   | 3.9   | 3.7   | 3.5   | 3.8   | 3.8   | 3.9   | 4.0  |
| Precipitation                                                                    | PrecAvg     | 1191   | 94    | 91    | 106   | 74    | 102   | 106   | 124   | 122   | 120   | 80    | 78    | 90   |
|                                                                                  | PrecMax     | 1727   | 221   | 150   | 278   | 213   | 220   | 266   | 268   | 249   | 394   | 306   | 169   | 194  |
|                                                                                  | PrecMin     | 718    | 18    | 6     | 27    | 12    | 11    | 14    | 31    | 15    | 14    | 9     | 7     | 11   |
|                                                                                  | PrecStd     | 253    | 38    | 35    | 52    | 43    | 51    | 63    | 60    | 63    | 89    | 57    | 46    | 46   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 22.0  | 23.5  | 26.2  | 30.1  | 32.3  | 35.0  | 35.8  | 34.1  | 32.4  | 29.5  | 24.8  | 23.3 |
|                                                                                  |             | MCWB   | 17.3  | 17.4  | 17.2  | 20.1  | 22.9  | 24.2  | 24.7  | 25.0  | 23.6  | 22.1  | 18.9  | 18.1 |
|                                                                                  | 2%          | DB     | 18.8  | 20.2  | 23.2  | 27.7  | 30.2  | 32.8  | 33.9  | 32.7  | 30.7  | 27.4  | 22.3  | 20.2 |
|                                                                                  |             | MCWB   | 15.3  | 14.8  | 15.9  | 18.7  | 21.9  | 23.7  | 25.1  | 24.7  | 23.6  | 21.1  | 18.1  | 17.2 |
|                                                                                  | 5%          | DB     | 16.4  | 17.6  | 20.9  | 25.8  | 28.5  | 31.4  | 32.6  | 31.4  | 28.9  | 25.2  | 20.1  | 17.8 |
|                                                                                  |             | MCWB   | 13.7  | 13.8  | 14.5  | 18.1  | 21.4  | 23.6  | 24.7  | 24.2  | 22.9  | 20.6  | 16.6  | 15.7 |
|                                                                                  | 10%         | DB     | 13.5  | 14.9  | 18.0  | 23.0  | 26.6  | 29.9  | 31.3  | 30.1  | 27.4  | 23.2  | 17.9  | 15.6 |
|                                                                                  |             | MCWB   | 10.8  | 10.9  | 14.0  | 17.2  | 20.4  | 23.0  | 24.4  | 24.0  | 22.4  | 19.4  | 15.1  | 13.4 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 18.8  | 18.7  | 19.2  | 22.2  | 24.6  | 26.2  | 27.4  | 27.3  | 26.0  | 24.6  | 21.1  | 20.2 |
|                                                                                  |             | MCDB   | 20.4  | 21.4  | 22.4  | 26.4  | 29.2  | 31.0  | 32.0  | 31.5  | 28.5  | 26.8  | 22.8  | 21.6 |
|                                                                                  | 2%          | WB     | 16.6  | 17.0  | 17.8  | 20.6  | 23.4  | 25.4  | 26.6  | 26.4  | 25.3  | 23.1  | 19.3  | 18.0 |
|                                                                                  |             | MCDB   | 18.6  | 19.2  | 20.9  | 25.1  | 27.8  | 30.2  | 31.2  | 30.1  | 28.0  | 25.5  | 21.0  | 19.5 |
|                                                                                  | 5%          | WB     | 14.2  | 14.4  | 16.5  | 19.4  | 22.5  | 24.8  | 25.8  | 25.7  | 24.5  | 21.8  | 17.5  | 16.1 |
|                                                                                  |             | MCDB   | 16.0  | 17.0  | 19.4  | 23.6  | 26.5  | 29.1  | 30.2  | 29.3  | 27.2  | 24.1  | 18.9  | 17.6 |
|                                                                                  | 10%         | WB     | 11.1  | 11.5  | 14.4  | 18.2  | 21.6  | 24.1  | 25.3  | 25.0  | 23.8  | 20.3  | 15.8  | 13.6 |
|                                                                                  |             | MCDB   | 13.4  | 14.3  | 17.9  | 22.2  | 25.1  | 27.9  | 29.4  | 28.5  | 26.3  | 22.7  | 17.7  | 15.4 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 8.8   | 9.4   | 9.5   | 10.1  | 9.1   | 8.6   | 8.4   | 8.0   | 7.9   | 9.1   | 9.5   | 8.9  |
|                                                                                  |             | MCDBR  | 12.6  | 13.6  | 13.3  | 12.7  | 11.2  | 10.4  | 9.9   | 9.4   | 9.3   | 10.4  | 11.3  | 11.1 |
|                                                                                  |             | MCWBR  | 9.7   | 9.6   | 7.9   | 6.0   | 5.1   | 4.1   | 3.7   | 3.7   | 4.0   | 5.4   | 7.4   | 8.8  |
|                                                                                  | 5% WB       | MCDBR  | 11.9  | 12.7  | 11.4  | 11.3  | 9.6   | 8.8   | 8.7   | 8.2   | 7.2   | 8.1   | 9.5   | 10.6 |
|                                                                                  |             | MCWBR  | 10.3  | 10.5  | 9.1   | 6.5   | 4.9   | 4.0   | 3.9   | 3.9   | 3.8   | 5.2   | 8.0   | 9.3  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.307  | 0.319 | 0.346 | 0.392 | 0.435 | 0.485 | 0.514 | 0.508 | 0.409 | 0.355 | 0.323 | 0.311 |      |
|                                                                                  | taud        | 2.542  | 2.504 | 2.460 | 2.341 | 2.253 | 2.165 | 2.098 | 2.100 | 2.366 | 2.516 | 2.553 | 2.563 |      |
|                                                                                  | Ebn at Noon | 889    | 919   | 920   | 889   | 852   | 807   | 779   | 774   | 842   | 866   | 862   | 860   |      |
|                                                                                  | Edh at Noon | 77     | 90    | 104   | 123   | 137   | 150   | 159   | 155   | 113   | 89    | 76    | 72    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 2.33   | 3.09  | 4.13  | 5.25  | 5.92  | 6.37  | 6.07  | 5.47  | 4.45  | 3.52  | 2.64  | 2.08  |      |
|                                                                                  | RadStd      | 0.19   | 0.33  | 0.33  | 0.38  | 0.52  | 0.42  | 0.29  | 0.30  | 0.37  | 0.40  | 0.22  | 0.16  |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|      |                         | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) | Station Only            | +0.39 | N/A     | -2.14  | N/A     | N/A   | N/A   | N/A         | N/A     | +154    | +107    |
| (47) | Regional (61 neighbors) | +0.46 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | +164    | +105    |

Nomenclature: See separate page